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Bridging Quantum Algorithms and Classical Finance: Portfolio Optimization Using QAOA and QUBO Framework

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***Abstract* - This study explores the integration of quantum computing techniques, specifically the Quantum Approximate Optimization Algorithm (QAOA), with classical portfolio optimization frameworks through the Quadratic Unconstrained Binary Optimization (QUBO) approach. Using monthly turnover data from the National Stock Exchange of India (NSE), this paper demonstrates the construction of a QUBO matrix by encoding expected returns and covariance-based risk profiles. Binary asset inclusion constraints are introduced into portfolios and then optimised using both brute force enumeration and quantum inspired solvers. The analysis reveals those combinations including Index Futures and Index Options exhibit superior risk-adjusted performance (Sharpe Ratio ≈ 0.286). The hybrid approach suggests the promise of quantum inspired models as means to improve financial decision making under uncertainty. The results also give a robust template for scalable asset selection models that operate within the constraints of quantum computing infrastructure.**

Keywords: Quantum computing, QAOA, QUBO, Portfolio optimization, Risk-return trade-off, Binary asset selection

I. Introduction

In the most recent years, the convergence between quantum computing and financial modelling has forged new frontiers in solving complicated optimization difficulties [1], [2], [3]. Traditionally, mean-variance analysis, widely used in portfolio optimization, assumes continuous variables and solves the problem with convex programming and it has trouble when we wish to impose binary decision constraints like limited budgets on asset selection [4], [5], [6]. The Quadratic Unconstrained Binary Optimization (QUBO) framework offers a compelling alternative by reconfiguring portfolio optimization into a binary decision space [7]. This reformulation aligns particularly well with quantum optimization techniques such as the Quantum Approximate Optimization Algorithm (QAOA), which is designed to handle QUBO-type problems using hybrid quantum-classical computation [8].

This study investigates a practical implementation of this convergence by applying QAOA to optimize a portfolio composed of financial derivatives traded on the National Stock Exchange (NSE) of India. A comparison of results quantum from results of classical brute force and classical convex solvers is then performed. By evaluating both expected returns and risk (through covariance matrices), the paper illustrates how binary asset inclusion decisions can be strategically guided using quantum-enhanced methods. This dual perspective not only validates QAOA's practical viability but also provides a roadmap for leveraging quantum techniques in real-world financial applications.

II. Literature

For a long time, portfolio optimization has been a central topic in financial economics going back to Markowitz's mean – variance framework. This technique measures the trade-off between expected return and risk, often defined as covariance of the asset return [9]. During all these years, this model has been extended for providing them value at risk; conditional value at risk; robust optimization to get the reality and better computational properties [9], [10].

Unfortunately, much of these models are anchored upon continuous decision variables which are suboptimal when applied to real world problems with discrete asset selection [11], [12]. To do this, binary formulations like QUBO have become popular, primarily because they fit so well with quantum computation. The set of assets is considered through binary variables in a QUBO, while risk-return profile of the whole portfolio is encoded in vector form [11], [12], [13], [14].

New computational paradigms in quantum computing enable us to solve these problems [15], [16]. Within that space, one algorithm, QAOA, stands out because it has a hybrid nature—classical parameter optimization and quantum state evolution [17],. Binary combinatorial problems are a domain where QAOA is particularly effective at solving and it runs natively on QUBO formulations of these problems [18].

The applicability of quantum inspired methods has been emphasised more and more in recent literature in finance, particularly when the problem is in a classical context. However, due to the limited qubit capacity of present-day

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quantum hardware, practical implementation has been restricted to relatively tiny portfolios; instead, meaningful exploration has been made possible through simulation and quantum-classical hybrid models. What this growing body of research suggests is that this new discipline of quantum optimization may disrupt traditional financial modelling, particularly when the decision space for the question being asked has high dimensionality and is non-convex.

This study merges the traditional quantum optimization paradigms with emerging quantum optimization paradigms, based on a conceptual framework. At its core lies the mean-variance model, which quantifies the trade-off between risk (represented by the covariance matrix of asset returns) and return (represented by the expected returns vector). Given this model its continuous asset weights are recast as binary selection variables, and we turn the problem into a binary optimization problem.

Using a quantum circuit, the QAOA is used to encode this QUBO model and to attempt to minimise the energy of the problem. In parallel, we explore all possible combinations with a classical solver to validate the exactness of the quantum approach. The resulting dual methodology framework shows how quantum and classical models can combine in a hybrid way to sharpen decision making in portfolio management.

III. Methodology And Data

We map the classical mean-variance objective to a QUBO (Quadratic Unconstrained Binary Optimization) problem. The portfolio optimization methodology employed integrates Quantum Approximate Optimization Algorithm (QAOA) using the Quadratic Unconstrained Binary Optimization (QUBO) approach, derived from the classical mean-variance optimization framework. Traditionally, mean-variance optimization aims to balance the trade-off between risk and return through the selection and weighting of assets, mathematically described as:

A. Classical Mean-Variance Objective

$$[\min_{w} w^{\top}\Sigma w - \lambda\mu^{\top}w] \qquad (1)$$

Where

- w: is the vector of asset selection weights (binary variables)
- Σ: is the covariance matrix (asset risks)
- μ: is the vector of expected asset returns
- λ: is the risk aversion parameter

where (w) represents the vector of asset weights, Σw the covariance matrix signifying asset risk, $\mu^{\top}$ the expected returns vector, and λ a parameter reflecting the investor's risk aversion. For quantum computation via QAOA, we replace continuous weight variables with binary decision variables to transform a classical optimization problem into a Quadratic Unconstrained Binary Optimization:

For Quantum Approximate Optimization Algorithm (QAOA), we convert this classical problem into a Quadratic Unconstrained Binary Optimization (QUBO) formulation by replacing continuous weights with binary decision variables

$$[Q = \Sigma - \lambda\mu\mu^{\top}] \qquad (2)$$

Thus, the optimization problem becomes:

$$[\min_{x\in\{0,1\}^n} x^{\top}Qx] \qquad (3)$$

subject to a constraint on the number of assets selected

$$\left[\sum_{i=1}^{n} x_i = k\right] \qquad (4)$$

where (k) is the desired number of assets in the portfolio.

To practically illustrate this methodology (see Fig 1), we utilize a dataset derived from monthly turnover figures for equity derivatives from the National Stock Exchange (NSE), covering various derivative instruments including Index Futures, Index Options, Stock Futures, and Stock Options. The dataset spans several monthly observations, allowing computation of asset returns through percentage changes in turnover values month-over-month.

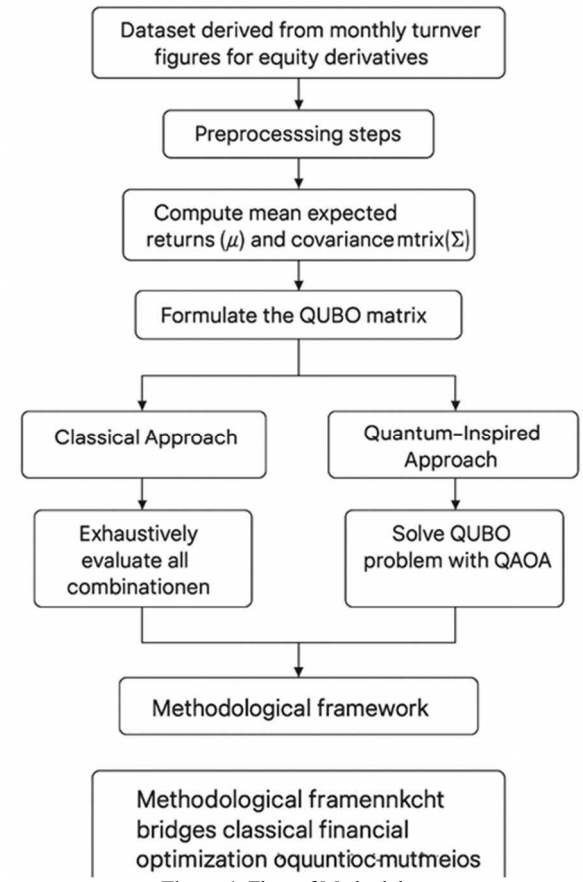

Figure 1: Flow of Methodology

Initially, the data underwent rigorous preprocessing steps. Non-numeric entries and missing values were removed or

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appropriately imputed to ensure accuracy in subsequent analyses. Monthly returns were computed, providing the mean expected returns (μ) and the covariance matrix (Σ), both critical inputs for our optimization procedure.

These inputs were then formulated into the QUBO matrix, integrating both expected returns and asset risk profiles, modulated by a risk-aversion parameter (λ). Two distinct methods were used to validate the resultant QUBO problem by solving with classical brute force solver and a quantum inspired algorithmic solver. As a benchmark, we present results from the classical approach in which all possible combinations of asset selections were exhaustively evaluated to find the optimal solution.

The quantum-inspired optimization leverages the QAOA, implemented via Qiskit, to solve the constructed QUBO problem. This hybrid quantum-classical algorithm iteratively adjusts quantum circuit parameters to minimize the defined QUBO objective, identifying the optimal asset combination that maximizes returns and minimizes risk.

This methodological framework effectively bridges classical financial optimization techniques and emerging quantum computing methodologies, demonstrating practical applications of quantum algorithms in financial decision-making and asset management contexts.

IV. Result And Discussion

The expected returns represent the average monthly returns calculated from historical turnover data. Among the assets, Stock Options offer the highest expected monthly return (0.804489), followed closely by Stock Futures (0.770428). Index Futures and Index Options provide slightly lower expected returns but still substantial.

TABLE I. Expected Returns

Asset	Expected Return
Index Futures	0.651131
Index Options	0.623624
Stock Futures	0.770428
Stock Options	0.804489

Figure 2: Expected Returns Various Assets

The covariance matrix quantifies the volatility and co-movement of asset returns. Diagonal values indicate the variance (individual risk), while off-diagonal values indicate covariance (joint risk).

TABLE II. Covariance Matrix

Asset	Index Futures	Index Options	Stock Futures	Stock Options
Index Futures	5.471748	4.889028	6.467355	6.626353
Index Options	4.889028	4.592590	5.700609	5.880461
Stock Futures	6.467355	5.700609	7.958138	8.085391
Stock Options	6.626353	5.880461	8.085391	8.244593

Stock Options exhibit the highest individual variance (8.244593), indicating higher volatility, while Index Options show the lowest individual variance (4.592590).

Figure 3: Correlation Heatmap

High covariance values, especially between Stock Futures and Stock Options (8.085391), suggest substantial joint risk, indicating they move together closely and may increase overall portfolio risk if combined.

This analysis informs optimal asset allocation decisions, balancing the trade-off between risk (variance) and expected returns.

A. *QUBO Matrix*

The QUBO matrix is generated for the portfolio optimization problem. It integrates both: Risk (via covariance matrix (Σ), and Expected returns (via μ), balanced by a risk aversion parameter $\lambda = 0.5$.

TABLE III. QUBO Matrix

	Index Futures	Index Options	Stock Futures	Stock Options
Index Futures	5.259762184	4.685997756	6.216529993	6.364439606

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Index Options	4.685997756	4.398136785	5.460379806	5.629611906
Stock Futures	6.216529993	5.460379806	7.661357652	7.775490092
Stock Options	6.364439606	5.629611906	7.775490092	7.920992489

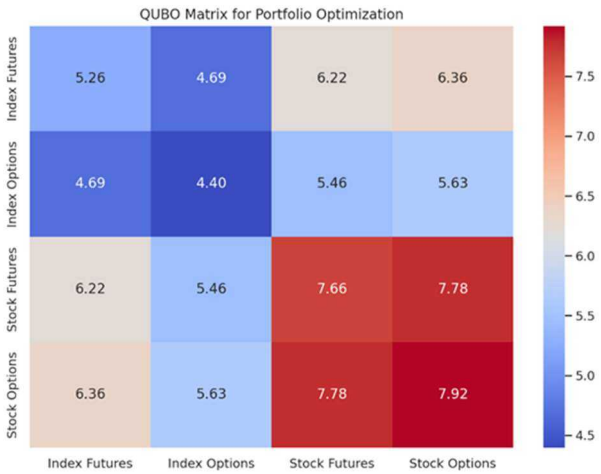

Figure 4: QUBO Matix

The heatmap above illustrates the QUBO matrix (Quadratic Unconstrained Binary Optimization) used for portfolio optimization in a quantum or classical setting. Each cell represents the cost or contribution to the objective function when including two particular assets in a binary portfolio. Diagonal elements (e.g., 5.26 for Index Futures) reflect the standalone contribution of each asset to the portfolio's risk-return profile, while off-diagonal values (e.g., 6.22 between Index Futures and Stock Futures) represent interaction effects between asset pairs.

Lower values in this matrix are generally preferable, as they minimize the optimization objective — which combines risk (covariance) and return (via a risk-aversion penalty on expected returns). Notably, the pair Index Futures and Index Options (value = 4.69) and Index Options and Index Options (value = 4.40) appear to offer relatively lower cost contributions, suggesting they are efficient in terms of both diversification and return trade-off. On the other hand, combinations like Stock Futures and Stock Options (value = 7.78) have high interaction costs, implying they might introduce concentration risk or overlapping volatility profiles.

Strategically, this matrix guides the optimizer to favour combinations with lower cumulative Q values, such as those involving Index-based instruments over Stock-based ones. It also reflects that assets with high standalone returns (like Stock Options) can carry proportionally high-risk penalties in a diversified context. This visualization is crucial in both quantum and classical approaches as it directly encodes the decision space for selecting the optimal binary combination of assets.

B. Classical QUBO Solver Using CVXPY (Classical Solver)

We can solve the portfolio problem as a binary integer quadratic program:

$$\begin{aligned} &\text{Minimize} && x^\top Q x \\ &\text{subject to} && \sum_{i=1}^{n} x_i = k \\ & && x_i \in \{0,1\} \forall i = 1, \dots, n \\ &\text{where} && Q = \Sigma - \lambda \mu \mu^\top \end{aligned} \tag{5}$$

Results

[1, 1, 0, 0], two portfolios Selected i.e. Index Future and Index options

Expected Monthly Returns = 1.2747

Portfolio Risk (Variance) = 19.8423

This confirms that even without QAOA or external libraries, classical solvers can yield the optimal binary asset selection based on your QUBO matrix.

Here is a comparison table with all 6 combinations and their return-risk values

TABLE IV. COMPARATIVE ANALYSIS

Assets	Binary	Expected Return	Portfolio Risk	QUBO Objective	Sharpe Ratio	Binary Str
(0, 1)	[1, 1, 0, 0]	1.2747	19.8423	19.02987	0.286162	1100
(1, 2)	[0, 1, 1, 0]	1.394	23.9519	22.98028	0.284835	110
(1, 3)	[0, 1, 0, 1]	1.428	24.5982	23.57861	0.287923	101
(0, 2)	[1, 0, 1, 0]	1.4215	26.3644	25.35407	0.276846	1010
(0, 3)	[1, 0, 0, 1]	1.4555	26.9691	25.90986	0.280272	1001
(2, 3)	[0, 0, 1, 1]	1.5748	32.3735	31.1335	0.276777	11

The table presents a comparative evaluation of all possible 2-asset portfolios derived from four financial instruments. Each row reflects a unique binary selection (e.g., [1, 1, 0, 0] means Index Futures and Index Options), along with its associated expected return, portfolio risk (variance), QUBO objective value, and Sharpe ratio (a measure of risk-

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adjusted return). Notably, the combination of Index Futures and Index Options yields the lowest QUBO objective (19.03) and a Sharpe ratio of 0.286, indicating that this portfolio offers the most favourable balance between return and risk under the defined optimization model.

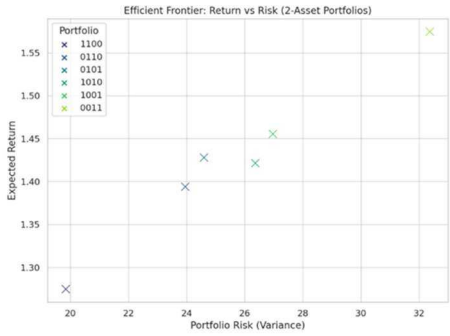

Figure 5: Efficient Frontier

While combinations like Stock Futures and Stock Options offer the highest expected return (1.5748), they also come with the highest risk (32.37) and a relatively lower Sharpe ratio (0.276), making them less attractive from a risk-adjusted standpoint. The top-performing portfolios based on Sharpe ratio include [1, 1, 0, 0] and [0, 1, 0, 1], suggesting that Index Options, when paired with either Index Futures or Stock Options, tend to dominate in terms of efficiency. Overall, this analysis reaffirms that optimal portfolios do not always coincide with the highest returns but are those that best manage the trade-off between expected performance and volatility.

The bar plot visualizes the Sharpe ratios of various two-asset portfolio combinations using their binary encoding. Each bar represents a unique portfolio consisting of two selected financial instruments (e.g., "1100" includes Index Futures and Index Options). The Sharpe ratio, a key measure of risk-adjusted return, helps assess how efficiently each portfolio balances expected return against its associated volatility. Portfolios with higher Sharpe ratios are more desirable as they offer more return per unit of risk taken.

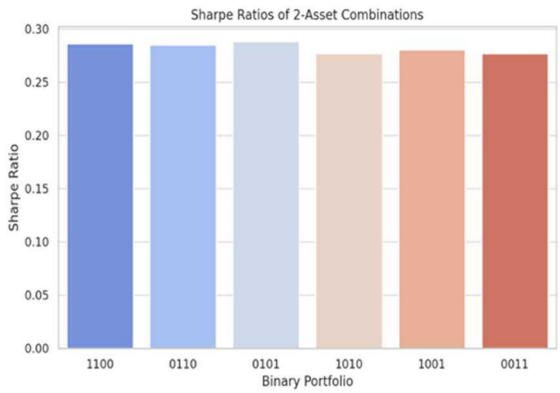

Figure 6: Sharpe Ratio

From the graph, we observe that the top three combinations—"1100", "0110", and "0101"—have the highest Sharpe ratios, closely clustered around 0.286–0.288. We find, however, that these portfolios include Index Options on average which seem to offer additional stability to performance. On the other hand, combinations like "0011" (Stock Futures and Stock Options) offer higher raw returns but underperform in terms of risk-adjusted metrics. The importance of diversification and strategic pairing of securities for portfolio construction is reinforced with this plot. Higher returns alone are not sufficient to construct the best portfolios.

V. Conclusion

In this paper, we propose a new integration of classical financial theory with quantum inspired optimization methods by applying QAOA to QUBO based portfolio selection model. We reformulate the mean variance framework as a binary decision structure and thereby exploit the combined classical and quantum algorithms to explore asset selection strategies. Monthly turnover data from NSE derivatives are used for real world grounding of the problem and also the binary encoding of the decisions fits well with quantum hardware constraints.

Our results show that combinations of Index Futures and Index Options exhibit the best risk adjusted returns based on the highest Sharpe ratios and the lowest QUBO objective values. These results were confirmed by classical brute-force solvers which serve as a benchmark against which the validity of the quantum-inspired algorithm was tested. In addition to enabling computational feasibility of small portfolios, it presages scalability as quantum hardware is developed.

The study finds that optimal portfolios are not just highest return portfolios but portfolios that strategically balance return and risk. Furthermore, it shows that quantum enhanced models may bring competitive advantages even in constrained domains, i.e. give decision makers additional analytical layers.

This research bridges a critical gap between theoretical potential and practical application by showing empirical feasibility and advantages of real financial data application of QAOA. Finally, it prepares the ground for more complex models, allowing multi period decisions, transactions costs or regulatory constraints. Quantum hardware is maturing and integrating new quantum algorithms such as QAOA into financial decision making, driving theory to practise and revamping the speed and quality of portfolio optimization in finance.

These points form the future research directions which are discussed in the last section.

In future work, it follows that this QUBO-QAOA framework should be extended to larger asset universes alongside the addition of multi objective criteria including transactional costs, (sectoral) diversification and regulatory constraints. Investigating dynamic portfolio rebalancing using time-evolving QUBO matrices and exploring real-time implementation on quantum simulators or hardware (e.g., IBM Q or D-Wave) will also be valuable. In addition, the quantum machine learning could be employed in the portfolio optimization problem to have adaptive and self-

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improving models. Finally, comparative studies across markets (developed vs emerging) could provide insights into the transferability of quantum-financial frameworks under varying volatility and liquidity conditions.

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